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SURFACE CLEANING APPARATUS

Abstract

A surface cleaning apparatus has an energy storage member unit with a first portion provided in the handle and a second portion provided in the main body of the apparatus.

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Background/Summary

FIELD

[0001] This disclosure relates generally to portable surface cleaning apparatuses such as hand vacuum cleaners.

INTRODUCTION

[0002] The following is not an admission that anything discussed below is part of the prior art or part of the common general knowledge of a person skilled in the art.

[0003] Various types of surface cleaning apparatus are known, including upright surface cleaning

apparatus, canister surface cleaning apparatus, stick surface cleaning apparatus, central vacuum systems, and hand carryable surface cleaning apparatus such as hand vacuum cleaners. Further, various designs for cyclonic surface cleaning apparatus, including battery operated cyclonic hand vacuum cleaners are known in the art.

SUMMARY

[0004] This summary is intended to introduce the reader to the more detailed description that follows and not to limit or define any claimed or as yet unclaimed invention. One or more inventions may reside in any combination or sub-combination of the elements or process steps disclosed in any part of this document including its claims and figures.

[0005] In accordance with one aspect of this disclosure, a hand vacuum cleaner which is operable using an energy storage member unit (e.g., a pack which contains one or more energy storage members such as one or more batteries, capacitors or the like). When the hand vacuum cleaner is ready for use, the energy storage member unit is at least partially located within the handle of the hand vacuum cleaner and in the main body of the hand vacuum cleaner. An advantage of this aspect is that larger batteries and/or more powerful batteries may be provided compactly within the vacuum cleaner while providing a hand vacuum cleaner with good weight distribution.

Accordingly, the vacuum cleaner may be more powerful and/or have a longer run time. Optionally, the energy storage member unit may be in the airflow path. Optionally, the energy storage member unit is removable.

[0006] In accordance with this aspect of this disclosure, there is provided a hand vacuum cleaner comprising: [0007] (a) an airflow path from a dirty air inlet positioned at a front end of the hand vacuum cleaner to a clean air outlet positioned rearward of the dirty air inlet; [0008] (b) an air treatment assembly comprising an air treatment chamber, the air treatment chamber positioned in the airflow path, the air treatment chamber has an air treatment chamber air inlet, an air treatment chamber air outlet, a front end, a rear end and a centrally positioned longitudinal axis extending between the front and rear ends of the air treatment chamber; [0009] (c) a main body housing a motor and fan assembly that is positioned in the air flow path, the motor and fan assembly having a motor axis of rotation; [0010] (d) a handle extending away from the main body; and, [0011] (e) an energy storage member unit, wherein a first portion of the energy storage member unit is provided in the handle and a second portion of the energy storage member unit is provided in the main body.

[0012] In accordance with this aspect of this disclosure, there is also provided a hand vacuum cleaner comprising: [0013] (a) an airflow path from a dirty air inlet positioned at a front end of the hand vacuum cleaner to a clean air outlet positioned rearward of the dirty air inlet; [0014] (b) an air treatment assembly comprising an air treatment chamber, the air treatment chamber positioned in the airflow path, the air treatment chamber has an air treatment chamber air inlet, an air treatment chamber air outlet, a front end, a rear end and a centrally positioned longitudinal axis extending between the front and rear ends of the air treatment chamber; [0015] (c) a main body housing a motor and fan assembly that is positioned in the air flow path, the motor and fan assembly having a motor axis of rotation; [0016] (d) a handle having a first end, an opposed second end, a hand grip portion positioned between the first end and the opposed second end of the handle and a handle axis that extends through the hand grip portion from the first end to the opposed second end; and, [0017] (e) an energy storage member unit comprising a plurality of energy storage members that have a length, which is a longest dimension of the energy storage member, and in use the length extends in a direction of the handle axis, [0018] wherein, in use, the motor axis of rotation intersects the energy storage member unit and the motor and fan assembly is located rearward of at least one of the energy storage members.

[0019] In accordance with this aspect sure, there is also provided a hand vacuum cleaning comprising: [0020] (a) an airflow path from a dirty air inlet positioned at a front end of the hand vacuum cleaner to a clean air outlet positioned rearward of the dirty air inlet; [0021] (b) an air treatment assembly comprising an air treatment chamber, the air treatment chamber positioned in

the airflow path, the air treatment chamber has an air treatment chamber air inlet, an air treatment chamber air outlet, a front end, a rear end and a longitudinal axis that extends between the front and rear ends of the air treatment chamber; [0022] (c) a main body housing a motor and fan assembly that is positioned in the air flow path, the motor and fan assembly having a motor axis of rotation; [0023] (d) a handle having a first end, an opposed second end, a hand grip portion positioned between the first end and the opposed second end of the handle and a handle axis that extends through the hand grip portion from the first end to the opposed second end; and, [0024] (e) an energy storage member unit comprising a plurality of energy storage members that have a length, which is a longest dimension of the energy storage member, and in use the length extends in a direction of the handle axis, [0025] wherein, in use, the longitudinal axis intersects the energy storage member unit and the motor and fan assembly is located rearward of at least one of the energy storage members.

[0026] In accordance with another aspect of the disclosure, a hand vacuum cleaner has a handle, which may be a pistol grip handle. An energy storage member unit may be insertable into the handle and optionally, also into the main body of the hand vacuum cleaner, through a lower end of the handle. The handle may have an open lower end and the energy storage member unit may optionally close the lower end of the handle. Alternately, the lower end of the handle may have an openable door which, when opened, enable the energy storage member unit to be installed or removed.

[0027] In accordance with this aspect, there is provided a hand vacuum cleaner comprising: [0028] (a) an airflow path from a dirty air inlet positioned at a front end of the hand vacuum cleaner to a clean air outlet positioned rearward of the dirty air inlet; [0029] (b) an air treatment assembly comprising an air treatment chamber, the air treatment chamber positioned in the airflow path, the air treatment chamber has an air treatment chamber air inlet, an air treatment chamber air outlet, a front end, a rear end and a centrally positioned longitudinal axis extending between the front and rear ends of the air treatment chamber; [0030] (c) a main body housing a motor and fan assembly that is positioned in the air flow path, the motor and fan assembly having a motor axis of rotation; [0031] (d) a handle extending away from the main body; and, [0032] (e) an energy storage member unit that is removably insertable into the handle.

[0033] It will be appreciated by a person skilled in the art that an apparatus or method disclosed herein may embody any one or more of the features contained herein and that the features may be used in any particular combination or sub-combination.

[0034] These and other aspects and features of various embodiments will be described in greater detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0035] For a better understanding of the described embodiments and to show more clearly how they may be carried into effect, reference will now be made, by way of example, to the accompanying drawings in which:

[0036] FIG. 1A is a side perspective view of a surface cleaning apparatus;

[0037] FIG. 1B is a cross-sectional side perspective view taken in a vertical plane extending through the lateral center of the cyclone air outlet of the surface cleaning apparatus of FIG. 1A;

[0038] FIG. 1C is a cross-sectional top perspective view taken in a horizontal plane that extends through the vertical center of the cyclone air outlet of the surface cleaning apparatus of FIG. 1A;

[0039] FIG. 2A is a cross-sectional side perspective view taken in a vertical plane extending through the lateral center of the cyclone air outlet of a further alternative embodiment of a surface cleaning apparatus;

[0040] FIG. 2B is a cross-sectional top perspective view taken in a horizontal plane that extends through the vertical center of the cyclone air outlet of the surface cleaning apparatus of FIG. 2A; [0041] FIG. 3A is a cross-sectional side perspective view taken in a vertical plane extending through the lateral center of the cyclone air outlet of a further alternative embodiment of a surface cleaning apparatus; [0042] FIG. 3B is a cross-sectional top perspective view taken in a horizontal plane that extends through the vertical center of the cyclone air outlet of the surface cleaning apparatus of FIG. 3A; [0043] FIG. 4A is a cross-sectional side perspective view taken in a vertical plane extending through the lateral center of the cyclone air outlet of a further alternative embodiment of a surface cleaning apparatus; [0044] FIG. 4B is a cross-sectional top perspective view taken in a horizontal plane that extends through the vertical center of the cyclone air outlet of the surface cleaning apparatus of FIG. 4A; [0045] FIG. 5A is a cross-sectional side perspective view taken in a vertical plane extending through the lateral center of the cyclone air outlet of a further alternative embodiment of a surface cleaning apparatus; [0046] FIG. 5B is a cross-sectional top perspective view taken in a horizontal plane that extends through the vertical center of the cyclone air outlet of the surface cleaning apparatus of FIG. 5A; [0047] FIG. 6 is a side perspective view of a further alternative embodiment of a surface cleaning apparatus, shown with the handle base and energy storage member unit removed; [0048] FIG. 7 is a side perspective view of a further alternative embodiment of a surface cleaning apparatus, shown with the handle base and energy storage member unit removed; [0049] FIG. 8 is a side perspective view of a further alternative embodiment of a surface cleaning apparatus, shown with the handle removed; and, [0050] FIG. 9 is a side perspective view of a further alternative embodiment of a surface cleaning apparatus, shown with the energy storage member unit removed. [0051] The drawings included herewith are for illustrating various examples of articles, methods, and apparatuses of the teaching of the present specification and are not intended to limit the scope of what is taught in any way.

DESCRIPTION OF VARIOUS EMBODIMENTS

[0052] Various apparatuses will be described below to provide an example of an embodiment of each claimed invention. No embodiment described below limits any claimed invention and any claimed invention may cover apparatuses that differ from those described below. The claimed inventions are not limited to apparatuses having all of the features of any one apparatus described below or to features common to multiple or all of the apparatuses described below. It is possible that an apparatus described below is not an embodiment of any claimed invention. Any invention disclosed in an apparatus described below that is not claimed in this document may be the subject matter of another protective instrument, for example, a continuing patent application, and the applicants, inventors or owners do not intend to abandon, disclaim or dedicate to the public any such invention by its disclosure in this document.

[0053] The terms “an embodiment”, “embodiment”, “embodiments”, “the embodiment”, “the embodiments”, “one or more embodiments”, “some embodiments”, and “one embodiment” mean “one or more (but not all) embodiments of the present invention(s)”, “unless expressly specified otherwise.

[0054] The terms “including”, “comprising”, and variations thereof mean “including but not limited to”, unless expressly specified otherwise. A listing of items does not imply that any or all of the items are mutually exclusive, unless expressly specified otherwise. The terms “a”, “an”, and “the” mean “one or more”, unless expressly specified otherwise.

[0055] As used herein and in the claims, two or more parts are said to be “coupled”, “connected”, “attached”, or “fastened” where the parts are joined or operate together either directly or indirectly (i.e., through one or more intermediate parts), so long as a link occurs. As used herein and in the

claims, two or more parts are said to be “directly coupled”, “directly connected”, “directly attached”, or “directly fastened” where the parts are connected in physical contact with each other. As used herein, two or more parts are said to be “rigidly coupled”, “rigidly connected”, “rigidly attached”, or “rigidly fastened” where the parts are coupled so as to move as one while maintaining a constant orientation relative to each other. None of the terms “coupled”, “connected”, “attached”, and “fastened” distinguish the manner in which two or more parts are joined together.

[0056] Some elements herein may be identified by a part number, which is composed of a base number followed by an alphabetical or subscript-numerical suffix (e.g., **112a**, or **1121**). Multiple elements herein may be identified by part numbers that share a base number in common and that differ by their suffixes (e.g., **1121**, **1122**, and **1123**). All elements with a common base number may be referred to collectively or generically using the base number without a suffix (e.g., **112**).

[0057] It should be noted that terms of degree such as “substantially”, “about”, and “approximately” as used herein mean a reasonable amount of deviation of the modified term such that the end result is not significantly changed. These terms of degree may also be construed as including a deviation of the modified term, such as by 1%, 2%, 5% or 10%, for example, if this deviation does not negate the meaning of the term it modifies.

[0058] Furthermore, the recitation of numerical ranges by endpoints herein includes all numbers and fractions subsumed within that range (e.g., **1** to **5** includes 1, 1.5, 2, 2.75, 3, 3.90, 4, and 5). It is also to be understood that all numbers and fractions thereof are presumed to be modified by the term “about” which means a variation of up to a certain amount of the number to which reference is being made if the end result is not significantly changed, such as 1%, 2%, 5%, or 10%, for example.

General Description of a Surface Cleaning Apparatus

[0059] Referring to the Figures, exemplary embodiments of surface cleaning apparatuses **100** are shown. The following is a general discussion of surface cleaning apparatuses **100** which provides a basis for understanding several of the features which are discussed herein. As discussed subsequently, each of the features may be used individually or in any particular combination or sub-combination in this or in other embodiments disclosed herein.

[0060] In the embodiments illustrated, the surface cleaning apparatus **100** is a hand-held vacuum cleaner, which is commonly referred to as a “hand vacuum cleaner” or a “handvac”. As used herein, a hand-held vacuum cleaner or hand vacuum cleaner or handvac is a vacuum cleaner that can be operated generally one-handedly to clean a surface while its weight is held by the same one hand. This is contrasted with upright and canister vacuum cleaners, the weight of which is supported by a surface (e.g. floor below) during use. Optionally, surface cleaning apparatus **100** could be removably mountable on a base so as to form part of, for example, an upright vacuum cleaner, a canister vacuum cleaner, a stick vac, a wet-dry vacuum cleaner and the like.

[0061] It will also be appreciated that a surface cleaning apparatus may use any configuration of the operating components and the airflow paths exemplified herein.

[0062] Referring to FIG. **1A**, a hand surface cleaning apparatus is exemplified as a hand vacuum cleaner with an apparatus front end **104**, an apparatus rear end **108**, an apparatus upper end **112** and an apparatus lower end **116**. The surface cleaning apparatus **100** includes a main body **120**. The main body **120** includes a main body housing **124**.

[0063] An apparatus air flow path **128** (see, e.g., FIGS. **1B-1C**) extends from an apparatus dirty air inlet **132** to an apparatus clean air outlet **136**. The apparatus dirty air inlet **132** may be provided at the apparatus front end **104**. The apparatus dirty air inlet **132** may be at an uppermost part of the front end **104** as exemplified, and may be directed forwardly as exemplified (i.e., opening forwardly). It will be appreciated that the dirty air inlet **132** may be located elsewhere, such as a lowermost part of the apparatus front end **104**.

[0064] The apparatus clean air outlet **136** is located rearward of the dirty air inlet **132** and may be provided at the apparatus upper end **112**. The apparatus clean air outlet **136** may be at a rearmost

part of the upper end **112**. It will be appreciated that the clean air outlet **136** may be located elsewhere, such as at other locations along the upper end **112**, or at the apparatus lower or rear ends **116**, **108**. It will also be appreciated that clean air outlet **136** may include multiple openings at different locations (e.g., on each lateral side of the surface cleaning apparatus **100**). As exemplified in FIG. 1A, the clean air outlet **136** is provided as part of the main body **120** and includes a grill **140**. The clean air outlet **136** may be in other configurations, shapes, and/or positions in other embodiments.

[0065] Referring to FIG. 1B, the apparatus dirty air inlet **132** may be provided at an inlet end **144** of an inlet conduit **148**. The inlet conduit **148** extends from the apparatus dirty air inlet **132** rearwardly to an inlet conduit outlet end **152**. Optionally, as exemplified, the inlet conduit outlet end **152** is within an internal air treatment chamber of the surface cleaning apparatus **100** (e.g., an air treatment chamber **180** which may be a cyclone chamber, described further below). Accordingly, the inlet conduit **148** may be position in the air treatment chamber **180**. Alternatively, the inlet conduit **148** may be provided external to the internal chamber (e.g., the inlet conduit may extend rearwardly along an upper or lower end of the air treatment chamber **180** and may have an outlet port that is also an inlet port provided in a sidewall (e.g., a sidewall) of the air treatment chamber **180**).

[0066] It will also be appreciated that, in some embodiments, the surface cleaning apparatus **100** may not include an inlet conduit **148**, and the apparatus dirty air inlet **132** may instead open directly into a downstream chamber (e.g., the air treatment chamber **180**) rather than being at an upstream end of a conduit.

[0067] The inlet conduit **148** may be a generally linear conduit having an inlet conduit longitudinal axis **156** along a longest dimension of the inlet conduit and extending between the inlet conduit inlet end **144** and the inlet conduit outlet end **152**. The inlet conduit longitudinal axis **156** may be generally horizontal when the apparatus upper end **112** is above the apparatus lower end **116**.

[0068] As exemplified, the inlet conduit **148** may form a nozzle **160** of the surface cleaning apparatus **100**. Alternatively, or additionally, the inlet conduit **148** may be connected or directly connected to an accessory. The accessory may be any suitable accessory tool such as a rigid air flow conduit which may include or be connectable to a surface cleaning head to form a stick vacuum cleaner, a crevice tool, a mini brush, and the like. The accessory may be coupled to the surface cleaning apparatus **100** such that the accessory is in air flow communication with the dirty air inlet **132** (e.g., in air flow communication with the inlet conduit **148**). For example, the accessory may be or include a conduit, and the conduit of the accessory may be received within the inlet conduit **148** or may receive the inlet conduit **148** within the accessory conduit. Optionally, one or more releasable fasteners may be used to couple the accessory to the surface cleaning apparatus **100**, such as inter-engagement members or magnets. Alternatively, or additionally, the accessory may be held in air flow communication with the dirty air inlet **132** via a friction fit (e.g., between an outer diameter of an accessory conduit and an inner diameter of the inlet conduit **148**, or vice versa).

[0069] As exemplified herein, the surface cleaning apparatus **100** includes an air treatment assembly **172**. The apparatus air flow path **128** extends through the air treatment assembly **172**. The air treatment assembly **172** is configured to remove particles of dirt and other debris from the airflow and/or otherwise treat the airflow. The air treatment assembly **172** includes one or more air treatment members **176**. Any air treatment member or members known in the art may be used. For example, the surface cleaning apparatus may use one or more cyclones, non-cyclonic momentum separators, bags, screens, physical filter media (e.g., foam, felt, HEPA) or the like.

[0070] It will be appreciated that the air treatment assembly **172** may include any suitable number of air treatment stages, each with any suitable number of air treatment members in parallel. For example, the air treatment member **176** may include a first separation stage with a single air treatment member or chamber (e.g., a cyclone or non-cyclonic momentum separator) and a

downstream second stage comprising a plurality of air treatment members or chambers (e.g., an array of cyclones in parallel with one another). As exemplified herein, the air treatment assembly **172** may comprise a single cleaning stage. A single cleaning stage may include, e.g., a single cyclonic stage (which may comprise one or more cyclones in parallel), or a single non-cyclonic momentum separator chamber. As exemplified, the single cleaning stage has a single air treatment member which is a single cyclone chamber, and, as such, the air treatment assembly **172** may be referred to as air treatment member **176**.

[0071] The exemplary air treatment member **176** includes an air treatment chamber **180**. The air treatment chamber **180** has a chamber housing **184** having a chamber first end wall **188** and a chamber second end wall **192** axially spaced from the chamber first end wall **188** along a centrally positioned chamber longitudinal axis **196**. The chamber first end wall **188** and chamber second end wall **192** may close the ends of the air treatment chamber **180**. The chamber first end wall **188** may be or comprise a front end of the chamber **180** and the chamber second end wall **196** may be or comprise a rear end of the chamber **180**. The chamber housing **184** also has a chamber sidewall **194** that extends between the chamber first end wall **188** and chamber second end wall **192**. The air treatment chamber **180** may be a cyclone chamber, as exemplified, and the chamber longitudinal axis **196** may be a cyclone axis of rotation. The chamber longitudinal axis **196** may be a generally horizontal axis when the apparatus upper end **112** is above the apparatus lower end **116** and may extend between the apparatus front end **104** and the apparatus rear end **108**.

[0072] An air treatment member longitudinal axis **200** extends between an air treatment member first end **204** (e.g., front end) and an air treatment member second end **208** (e.g., rear end). As exemplified, the air treatment member longitudinal axis **200** may be the same as the chamber longitudinal axis **196**. As exemplified, the air treatment member longitudinal axis **200** may be centrally located within the air treatment chamber **180**. The air treatment member longitudinal axis **200** may be generally horizontal when the apparatus upper end **112** is above the apparatus lower end **116**.

[0073] The air treatment chamber **180** has an air treatment chamber air inlet **212** including a chamber inlet opening **216** and an air treatment chamber air outlet **220** including a chamber outlet opening or port **224**. The apparatus air flow path **128** extends through the chamber inlet opening **216** and the chamber outlet opening or port **224**. However, it will be appreciated that in some embodiments, the air treatment chamber air inlet **212** and/or the air treatment chamber air outlet **220** may include more than just an opening and may be at any location.

[0074] Optionally, as exemplified, the air treatment chamber air outlet **220** may include a screen which, if the chamber **180** is a cyclone, may be a vortex finder **228**. The vortex finder **228** may overlie the air treatment chamber air outlet **220**. As shown, the vortex finder **228** may extend inwardly into the air treatment chamber **180**.

[0075] As exemplified, the air treatment assembly **172** may include a dirt collection region **232**. As exemplified in FIG. **1B**, the dirt collection region **232** may be external to the air treatment chamber **180** in a separate dirt collection chamber **236**. The separate dirt collection chamber **236** may communicate with the air treatment chamber **180** via one or more dirt outlets **240** (e.g., an opening in a wall of the air treatment chamber **180** or a gap between walls of the air treatment chamber **180**). Alternatively, the dirt collection region **232** may be an area of the air treatment chamber **180** (i.e., is internal to the air treatment chamber **180**). The dirt collection region **232** may include, e.g., a forward portion of the air treatment chamber **180**.

[0076] It will be appreciated that the air treatment assembly **172** may include any suitable number of discrete dirt collection regions **232**. In some embodiments, the air treatment assembly **172** may include separate collection regions for fine dirt and for coarse dirt, or separate dirt outlets to a common dirt collection chamber for fine dirt and for coarse dirt. It will be understood that the air treatment chamber **180** and dirt collection region **232** may be of any configuration suitable for separating dirt from an air stream and collecting the separated dirt, respectively.

[0077] It will also be appreciated that the air treatment chamber **180** and/or dirt collection chamber **236** may be openable to, for example, allow for dirt removal. For example, part or all of chamber first end wall **188**, chamber sidewall **194** or rear end wall **192** may be openable. For example, as shown in FIG. **1B**, a portion of the chamber sidewall **194** is openable downwardly by pivoting the portion about a pivot joint **242**.

[0078] A suction motor and fan assembly, which may be a suction motor **164**, is provided to generate air flow through the air flow path **128**, and is positioned within a motor housing **168**. It will be appreciated that, instead of a clean air motor which is exemplified, a dirty air motor may be used

[0079] The suction motor housing **168** may form part of the outer surface of the main body housing **124**, or may be internal thereto. The suction motor housing **168** may be of any suitable construction, including any of those exemplified herein.

[0080] Optionally, as exemplified, the suction motor **164** is a clean air motor that is positioned downstream from the air treatment member **176** and rearward thereof. The suction motor **164** may be located at the apparatus rear end **108** as exemplified or at any location between the air treatment member **176** and the rear end **108**. In addition, the suction motor **164** may be provided at any vertical elevation within the main body. As exemplified, the suction motor **164** may be located at the apparatus lower end **116**. In other embodiments, the suction motor **164** may be located at the apparatus upper end **112**. Air may travel rearwardly from the air treatment assembly **172** to the suction motor **164**, and air flow direction between the air treatment member **176** and the suction motor **164** may have a rearward component at each point along the way.

[0081] The suction motor **164** rotates about a central motor axis of rotation **244**. Optionally, when the apparatus upper end **112** is positioned above the apparatus lower end **116**, the motor axis of rotation **244** is oriented generally horizontally and extends between the apparatus front end **104** and the apparatus rear end **108**. In other examples, however, the motor axis of rotation **244** may extend at any angle to the horizontal, or it may extend vertically. Accordingly, the suction motor **244** may be oriented in any direction within the surface cleaning apparatus **100**. Optionally, as exemplified, the motor axis of rotation **222** extends in a common direction with the cyclone axis of rotation and, optionally, they may be coaxial.

[0082] Optionally, one or more pre-motor filters **248** may be placed in the air flow path **128** downstream of the air treatment member **176** and upstream of the suction motor **164**. Alternatively, or in addition, one or more post-motor filters (not shown) may be provided downstream from the suction motor **124** and upstream of the clean air outlet. The pre-motor filter **248** and the post-motor filter may be formed from any suitable physical, porous filter media and may have any suitable shape, including the examples disclosed herein. For example, the pre-motor filter **248** and/or the post-motor filter may be one or more of a foam filter, felt filter, HEPA filter, other physical filter media, electrostatic filter, and the like. Optionally, one or both of the pre-motor filter **248** and the post-motor filter includes a series of filter media, and, optionally, each downstream filter media has finer pores than the preceding upstream screen.

[0083] The pre-motor filter **248** may have any suitable shape and may be at any location. As exemplified, the pre-motor filter **248** may be configured as a generally cylindrical or frusto-conical foam filter with a hollow, open interior (e.g., FIG. **1B**). Alternatively, the pre-motor filter **248** may be a generally flat, slab-like filter. The pre-motor filter **248** may have a longest dimension in a longitudinal (horizontal) direction along a pre-motor filter longitudinal axis **252**. The pre-motor filter longitudinal axis **252** may be generally parallel to and/or coaxial with one or more of the air treatment member longitudinal axis **200**, the motor axis of rotation **244**, and the inlet conduit longitudinal axis **156**.

[0084] The pre-motor filter **248** may be provided in a pre-motor filter housing **256**. The pre-motor filter housing **256** may be of any suitable construction, including any of those exemplified herein. The pre-motor filter housing **256** may be openable or accessible to allow the pre-motor filter **248** to

be cleaned and/or replaced. Optionally, the pre-motor filter **248** may be located at the rear end of the air treatment assembly and, if the air treatment assembly is removable, the pre-motor filter **248** may be accessible when the air treatment assembly is removed (e.g., an upstream face of a pre-motor filter seated in the main body may be revealed or accessible when the air treatment assembly is removed) or the downstream face of the pre-motor filter may be revealed or accessible (e.g., the pre-motor filter is seated in a rear end of the removable air treatment assembly) when the air treatment member is removed. Similarly, if a rear end of the air treatment assembly is pivotally mounted, e.g., to the main body, a face of the pre-motor filter may be revealed or accessible when the air treatment assembly is pivoted open.

[0085] Optionally, a post-motor filter (not shown) may be placed in the air flow path **128** downstream of the suction motor **164** and upstream of apparatus clean air outlet **136**. The post-motor filter may be provided in a post-motor filter housing, which may be part of the main body. The post-motor filter housing may be of any suitable construction. The post-motor filter may have various configurations, shapes and/or positions. For example, the post-motor filter may be located radially outwards of the suction motor **164**. The post-motor filter may alternatively be sandwiched between the motor housing **168** and the main body housing **124**.

[0086] In the embodiments disclosed herein, the surface cleaning apparatus **100** has a handle **260** (e.g., FIG. 1A). Handle **260** may extend away from the main body **120**, e.g., it may extend outwardly from the apparatus lower end **116**. It will be appreciated that the main body housing **124** and/or handle **260** may be in other configurations, shapes, and/or positions in other embodiments. For example, in embodiments wherein the energy storage member unit does not extend into the main body, the handle may be the rear end of the hand vacuum cleaner and connected to the main body by one or more forwardly extending bridge portions.

[0087] As exemplified, the handle **260** may be a pistol grip handle with a hand grip portion **264** that extends generally vertically. The handle **260** has a longest dimension in the direction of a handle longitudinal axis **268**, which extends through a hand grip portion **264** from the handle upper end **272** to the handle lower end **276**. The handle upper end **272** of the pistol grip handle may be mounted to a lower end of the surface cleaning apparatus **100**.

[0088] The handle may include a handle base **280** (e.g., FIG. 2A) for acting as a stand when setting down the surface cleaning apparatus **100** on a surface.

[0089] The handle may underlie one or more of the suction motor **164**, the pre-motor filter **248**, and/or the air treatment assembly **172**. Arranging the handle below a heavy and/or bulky component of the surface cleaning apparatus **100** may result in a more desirable hand-feel of the surface cleaning apparatus **100**.

[0090] Optionally, energy storage members **288** may also be provided in the handle base **280**.

Vertical Positioning of the Energy Storage Member Unit

The following is a description of the vertical positioning of energy storage member unit **284** that may be used by itself in any surface cleaning apparatus or in any combination or sub-combination with any other feature or features described herein.

[0091] In accordance with this aspect, power is supplied to the surface cleaning apparatus **100** at least one energy storage member unit **284** and optionally only a single energy storage member unit **284** (e.g., FIG. 1B). The energy storage member unit **284** may comprise a plurality of energy storage members **288**, which may be a plurality of batteries or capacitors or the like.

[0092] The energy storage member unit **284** may be a single body which contains a plurality of energy storage members. The energy storage members may be arranged in a single row or a plurality of rows. As exemplified in FIGS. 1B and 1C, an energy storage member unit **284** has two rows of batteries wherein the long dimension of the batteries extend in a long dimension of the energy storage member unit **284**. Accordingly, then installed in the handle, the axis of the energy storage member unit **284** and the longitudinal axis of the batteries therein may be parallel to the handle axis.

[0093] It will be appreciated that the energy storage member unit **284** may be dimensioned so as to fit between a user's fingers and palm when the user places their hand around the handle and grips the handle. Accordingly, as exemplified in FIGS. **1B** and **1C**, the energy storage member unit **284** has two rows of batteries, one behind the other. It will be appreciated that, in other embodiments, the rows may be positioned laterally side by side.

[0094] As exemplified in FIG. **1B**, the energy storage member unit **284** comprises a plurality of energy storage members **288** that, in use, are axially arranged in a direction of the handle axis **268**. Each energy storage member **288** has a length **300**. The length **300** is the longest dimension of the energy storage member **288**. In use, the length **300** extends in the direction of the handle axis **268**, as exemplified, for example in FIG. **1B**. As previously described, in some embodiments the energy storage members **288** may be a plurality of batteries **288**. Each battery **288** has a positive end **304** and a negative end **308**. In FIG. **1B**, the positive end **304** is depicted at an upper end of the battery **288**. It will be appreciated that in other embodiments, the positive end **304** may be located at a lower end of the battery **288**. A battery axis **312** extends between the positive end **304** and the negative end **308**. In use, the battery axis **312** extends in a common direction as the handle axis **268**. As exemplified, for example in FIG. **1B**, the batteries **288** may be stacked end to end within the handle **260** and extend into the main body **120**.

[0095] It will be appreciated that, in other embodiment, the battery axis may extend transversely (i.e., in a plane generally transverse to the cyclone axis of rotation).

[0096] In accordance with this aspect, at least a portion of the energy storage member unit **284** is located within both the handle **260** and the main body **120**. As exemplified in FIGS. **1B**, **2A**, **3A**, **4A** and **5A**, a first portion **292** of the energy storage member unit **284** is provided in the handle **260**, in the hand grip portion **264** as exemplified. A second portion **296** of the energy storage member unit **284** is provided in the main body **120**. Providing for the energy storage member unit **284** to extend from within the handle **260** into the main body **120** allows for more and/or larger energy storage members **288** to be used to power the surface cleaning apparatus **100** while maintaining a compact form. This may allow for the surface cleaning apparatus **100** to have a longer run time and/or stronger suction power.

[0097] It will be appreciated that the second portion **296** of the energy storage member unit **284** may extend from the handle **260** part wall into the main body or through most or all of the vertical height of the main body. As exemplified herein, in the orientation of FIG. **2A**, the surface cleaning apparatus **100** has a lower side **332** and an opposed upper side **336**. The handle **260** is provided on the lower side **332**. As exemplified in FIGS. **1B**, **2A**, and **3A**, the energy storage member unit **284** extends from the lower side **332** through all of the vertical height of the main body to the upper side **336**. Accordingly, the energy storage member unit **284** extends generally through an entire height **316** of the main body housing **124**. Therefore, the second portion **296** extends from the lower side **332** to the upper side **336**.

[0098] In other embodiments, such as in FIGS. **4A** and **5A**, the energy storage member unit **284** may not extend all the way from the lower side **332** to the upper side **336** (i.e., through the entire height **316**). As exemplified in FIGS. **4A** and **5A**, the second portion **296** of the energy storage member unit **284** extends only approximately one-quarter of the height **316**. In other embodiments, the second portion **296** may extend through any amount of the height **316** (e.g., one-half of height **316**, three-quarters of height **316**, etc.). In different embodiments, the amount the second portion **296** that extends into the main body **120** (i.e., the distance the second portion **296** extends away from the lower **332** towards the second rearwardly extending side **336**) may differ due to the use of energy storage members **288** of different dimensions.

[0099] It will be appreciated that, if the energy storage member unit **284** is shorter than the axial length of the handle **260**, then the lower end of the energy storage member unit **284**, in the orientation of FIG. **2A**, may be at an elevation higher than the lower end of the handle **260** and therefore the center of gravity of, e.g., the batteries, may be located closer to the elevation of the

suction motor. Accordingly, for example, in the embodiment of FIG. 2A, each row in the energy storage member unit **284** may consist of 2 batteries (i.e., the energy storage member unit **284** may have 4 batteries). In such a case, the lower two batteries shown in FIG. 2A are not provided and the lower end of the energy storage member unit **284** is located at about a mid-height of the handle **260** [0100] Accordingly, it will be appreciated that an axial projection of the suction motor may intersect part of the second portion **296** of the energy storage member unit **284**, e.g., the upper portion thereof in the embodiment of FIG. 4A. Alternately, in the embodiment of FIG. 2A, an axial projection of the suction motor may intersect all or essentially all of the second portion **296** of the energy storage member unit **284**. Optionally, if the motor axis extends in the forward/rearward direction, then the motor axis of rotation **224** may intersect the second portion **296**, e.g., an upper end thereof or a middle of the vertical height of the second portion **296**.

[0101] It will be appreciated that the second portion **296** of the energy storage member unit **284** may be positioned in the apparatus air flow path **128**. The air flow path **128** may flow over, around and/or above the second portion **296** of the energy storage member unit **284**. In some embodiments, the air flow path **128** may flow through the energy storage members **288** (e.g., between the batteries of a battery pack) or through air flow passages through, e.g., a battery pack. Airflow over the second portion **296** may help cool the energy storage members **288**.

[0102] It will be appreciated that the air exiting the air treatment chamber outlet **220** may still carry dust and/or debris. It may be desirable, therefore, to protect the second portion **296** of the energy storage member unit **284** that is in the air flow path **128** in order to prevent the accumulation of dust and/or debris on the energy storage members **288**. In order to protect the second portion **296**, various methods may be used to block the air flow path from contact with the second portion **296**. This may help prevent damage to the energy storage members **288** and extend their lifetime.

[0103] For example, in the embodiment shown in FIG. 2A, the energy storage member unit **284** is encased in a casing **320**. The casing **320** may cover all sides of the energy storage member unit **284** that is in the main body **120** such that the air flow is never in direct contact with the energy storage member unit **284**. Accordingly, for example, the energy storage member unit **284** may be a sealed enclosure, like a battery pack.

[0104] It will be appreciated that the casing **320** may be in other configurations, shapes, and/or positions in other embodiments.

[0105] In the embodiment shown in FIG. 2B, the surface cleaning apparatus **100** may include an energy storage protection wall **324**. The energy storage protection wall **324** may encircle the second portion **296** to prevent air from flowing over the second portion **296**. The energy storage protection wall **324** may force the airflow path **128** around and/or above the second portion **296**. The energy storage protection wall **324** may extend from the lower side **332** to the upper end of the second portion **296** (when installed) and optionally to the upper side **336**. The energy storage protection wall **324** may extend only partially from the lower side **332** towards the upper side **336**.

Accordingly, the energy storage protection wall **324** may be in the form of cup or container having a lower open side into which the second portion **296** is inserted. The energy storage protection wall **324** may be mounted or integrally formed to an inner surface of the lower side **332**. It will be appreciated that the energy storage protection wall **324** may be in other configurations, shapes and/or positions in other embodiments.

[0106] It will be appreciated that the energy storage member unit **284** may be removable from the surface cleaning apparatus **100** in any suitable manner. For example, the energy storage member unit **284** may be removably insertable into the handle **260** as discussed subsequently.

Axial Positioning of the Energy Storage Member Unit

[0107] The following is a description of the axial positioning of energy storage member unit **284** that may be used by itself in any surface cleaning apparatus or in any combination or sub-combination with any other feature or features described herein.

[0108] In accordance with this aspect, if a pre-motor filter is provided, then some or all of the

energy member unit **284** may be located forward or rearward of the pre-motor filter **248**.

[0109] As exemplified in FIGS. **1B**, **2A** and **4A**, the second portion **296** of the energy member unit **284** is be located forward of the pre-motor filter **248**. Alternately, as exemplified in FIGS. **3A** and **5A**, the second portion **296** of the energy member unit **284** is be located rearward of the pre-motor filter **248** and forward of the suction motor **164**.

[0110] In either case, the second portion **296** of the energy member unit **284** is located forward of the suction motor. Arranging the energy member unit **284** forward of the suction motor **164** may result in a more desirable weight distribution and hand-feel of the surface cleaning apparatus **100**, particularly if the second portion **296** is located rearward of the pre-motor filter **248** and, optionally adjacent a forward (air inlet end) of the suction motor **264**.

[0111] It will be appreciated that the energy storage member unit **284** may be removable from the surface cleaning apparatus **100** in any suitable manner. For example, the energy storage member unit **284** may be removably insertable into the handle **260** as discussed subsequently.

Removable Energy Storage Member Unit

[0112] The following is a description of the removability of energy storage member unit **284** that may be used by itself in any surface cleaning apparatus or in any combination or sub-combination with any other feature or features described herein.

[0113] Optionally, as discussed elsewhere herein, the energy storage member unit **284** may be removable. When installed, the energy storage member unit **284** may be located at least partially within the handle **260**. The energy storage member unit may have a lower end, in the orientation of FIG. **1B**, that is recessed inwardly from the lower end **276** of the handle **260** or at the handle lower end **276**.

[0114] In accordance with this aspect, the upper end of the energy storage member unit **284**, when installed, may be located in the handle or in the main body as discussed previously.

[0115] In accordance with this aspect, the energy storage member unit **284** is removably insertable into the handle **260**.

As exemplified in FIG. **6**, the energy storage member unit **284** may be insertable into the handle **260** through the handle lower end **276**. In this manner, the energy storage member unit **284** may be slidably received in the direction of the handle axis into the handle **260**. As shown in FIG. **6**, the handle **260** may remain mounted to the main body **120** when the energy storage member unit **284** is removed.

[0116] If the handle **260** has a handle base **280**, as exemplified in FIG. **6**, the handle base **280** may be attached to the energy storage member unit **284** such that the handle base **280** is removed from the handle **260** along with the energy storage member unit **284** and is positioned at the lower end of the handle when the energy storage member unit **284** is in the installed position as exemplified in FIG. **5A**.

[0117] The base may be fixedly mounted to the energy storage member unit **284** or it may be removably attachable to the energy storage member unit **284** and/or the lower end of the hand grip portion. For example, the base may have a cavity into which the lower end of the energy storage member unit **284** or the lower end of the hand grip portion is insertable. Alternately, the base may have a protrusion (see e.g., FIG. **7**) which is insertable into a cavity in the lower end of the energy storage member unit **284** or the lower end of the hand grip portion.

[0118] As exemplified in FIG. **7**, the handle base **280** is removable from the handle **260**. Therefore, the handle base **280** may be removed from the handle **260** prior to removing the energy storage member unit **284** through the lower end of the hand grip portion of the handle **260**.

[0119] As exemplified in FIG. **9**, the handle **260** has a handle pivot joint **328**. The handle base **280** is pivotably mounted to the handle **260** via the handle pivot joint **328**. The handle base **280** may be pivoted away from the lower end of the hand grip portion of the handle **260** to open the handle lower end **276** and allow for the energy storage member unit **284** to be removed from the handle **260**.

[0120] Alternately, as exemplified in FIG. 8, the handle **260** may be removably mounted to the remainder of the surface cleaning apparatus **100** (i.e., the main body **120**). The handle **260** is removable with the energy storage member unit **284**. In other words, the handle **260** containing the energy storage member unit **284** may be removed from the main body **120**. With the handle **260** removed from the main body **120**, the energy storage member unit **284** may subsequently be removed from the handle **260**. Alternately, the energy storage member unit may be permanently mounted in the handle **260** or it may form part or all of the handle **260**. It will be appreciated that if a handle base **280** is provided, the handle base may be permanently attached to the handle **260** or removable or pivotally mounted as discussed previously, e.g., to enable removal of the energy storage member unit **284** from the handle.

[0121] It will be appreciated that, if the handle is removable for the hand vacuum cleaner and the energy storage member unit **284** is removable from the handle, then the energy storage member unit **284** may be removeable through an open upper end of the hand after the handle has been removed from the hand vacuum cleaner.

[0122] It will also be appreciated if the energy storage member unit **284** has a second portion **296** that is received in the main body when the energy storage member unit **284** is installed in an operating position, then when the handle **260** is removed, the second portion **296** may extend upwardly out of the upper end of the handle and function as a grip portion for removing a removable energy storage member unit **284** upwardly out of the handle.

[0123] Accordingly, what has been described above is intended to be illustrative of the claimed concept and non-limiting. It will be understood by persons skilled in the art that other variants and modifications may be made without departing from the scope of the invention as defined in the claims appended hereto. The scope of the claims should not be limited by the preferred embodiments and examples, but should be given the broadest interpretation consistent with the description as a whole.

CLAUSES

Clause Set 1

[0124] 1. A hand vacuum cleaner comprising: [0125] (a) an airflow path from a dirty air inlet positioned at a front end of the hand vacuum cleaner to a clean air outlet positioned rearward of the dirty air inlet; [0126] (b) an air treatment assembly comprising an air treatment chamber, the air treatment chamber positioned in the airflow path, the air treatment chamber has an air treatment chamber air inlet, an air treatment chamber air outlet, a front end, a rear end and a centrally positioned longitudinal axis extending between the front and rear ends of the air treatment chamber; [0127] (c) a main body housing a motor and fan assembly that is positioned in the air flow path, the motor and fan assembly having a motor axis of rotation; [0128] (d) a handle extending away from the main body; and, [0129] (e) an energy storage member unit, wherein a first portion of the energy storage member unit is provided in the handle and a second portion of the energy storage member unit is provided in the main body.

[0130] 2. The hand vacuum cleaner of clause 1 wherein the handle has a first end, an opposed second end, a hand grip portion and a handle axis that extends through the hand grip portion from the first end to the opposed second end and, in use, the first portion of the energy storage member unit is provided in the hand grip portion.

[0131] 3. The hand vacuum cleaner of clause 2 wherein the handle is a pistol grip handle.

[0132] 4. The hand vacuum cleaner of clause 1 wherein the handle has a first end, an opposed second end, a hand grip portion and a handle axis that extends through the hand grip portion from the first end to the opposed second end, the energy storage member unit comprises a plurality of energy storage members that, in use, are axially arranged in a direction of the handle axis.

[0133] 5. The hand vacuum cleaner of clause aim 1 wherein the handle has a first end, an opposed second end, a hand grip portion and a handle axis that extends through the hand grip portion from the first end to the opposed second end, the energy storage member unit comprises a plurality of

energy storage members that have a length, which is a longest dimension of the energy storage member, and, in use, the length extends in a direction of the handle axis.

[0134] 6. The hand vacuum cleaner of clause 1 wherein the handle has a first end, an opposed second end, a hand grip portion and a handle axis that extends through the hand grip portion from the first end to the opposed second end, the energy storage member unit comprises a plurality of batteries that have a positive end and a negative end, a battery axis extends between the positive and negative ends and, in use, the battery axis extends in a common direction as the handle axis.

[0135] 7. The hand vacuum cleaner of clause 1 wherein, in use, the energy storage member unit is positioned in the air flow path.

[0136] 8. The hand vacuum cleaner of clause 1 wherein, in use, the second portion of the energy storage member unit is positioned forward of the motor and fan assembly.

[0137] 9. The hand vacuum cleaner of clause 1 wherein the main body further comprises a pre-motor filter and, in use, the second portion of the energy storage member unit is positioned forward of the motor and fan assembly and forward of the pre-motor filter.

[0138] 10. The hand vacuum cleaner of clause 1 wherein the main body further comprises a pre-motor filter and, in use, the second portion of the energy storage member unit is positioned forward of the motor and fan assembly and rearward of the pre-motor filter.

[0139] 11. The hand vacuum cleaner of clause 1 wherein the energy storage member unit is removable from the hand vacuum cleaner.

[0140] 12. The hand vacuum cleaner of clause 1 wherein the energy storage member unit is removable insertable into the handle.

[0141] 13. The hand vacuum cleaner of clause 1 wherein the handle has a first end provided on the main body, an opposed second end and a hand grip portion positioned between the first end and the opposed second end of the handle, and the energy storage member unit is insertable into the second end of the handle.

[0142] 14. The hand vacuum cleaner of clause 1 wherein, in use, the motor axis of rotation intersects the second portion of the energy storage member unit.

[0143] 15. A hand vacuum cleaner comprising: [0144] (a) an airflow path from a dirty air inlet positioned at a front end of the hand vacuum cleaner to a clean air outlet positioned rearward of the dirty air inlet; [0145] (b) an air treatment assembly comprising an air treatment chamber, the air treatment chamber positioned in the airflow path, the air treatment chamber has an air treatment chamber air inlet, an air treatment chamber air outlet, a front end, a rear end and a centrally positioned longitudinal axis extending between the front and rear ends of the air treatment chamber; [0146] (c) a main body housing a motor and fan assembly that is positioned in the air flow path, the motor and fan assembly having a motor axis of rotation; [0147] (d) a handle having a first end, an opposed second end, a hand grip portion positioned between the first end and the opposed second end of the handle and a handle axis that extends through the hand grip portion from the first end to the opposed second end; and, [0148] (e) an energy storage member unit comprising a plurality of energy storage members that have a length, which is a longest dimension of the energy storage member, and in use the length extends in a direction of the handle axis, [0149] wherein, in use, the motor axis of rotation intersects the energy storage member unit and the motor and fan assembly is located rearward of at least one of the energy storage members.

[0150] 16. The hand vacuum cleaner of clause 15 wherein, in use, a portion of the energy storage member unit is positioned in the main body.

[0151] 17. The hand vacuum cleaner of clause 15 wherein, in use, a portion of the energy storage member unit is positioned rearward of a pre-motor filter.

[0152] 18. A hand vacuum cleaner comprising: [0153] (a) an airflow path from a dirty air inlet positioned at a front end of the hand vacuum cleaner to a clean air outlet positioned rearward of the dirty air inlet; [0154] (b) an air treatment assembly comprising an air treatment chamber, the air treatment chamber positioned in the airflow path, the air treatment chamber has an air treatment

chamber air inlet, an air treatment chamber air outlet, a front end, a rear end and a longitudinal axis that extends between the front and rear ends of the air treatment chamber; [0155] (c) a main body housing a motor and fan assembly that is positioned in the air flow path, the motor and fan assembly having a motor axis of rotation; [0156] (d) a handle having a first end, an opposed second end, a hand grip portion positioned between the first end and the opposed second end of the handle and a handle axis that extends through the hand grip portion from the first end to the opposed second end; and, [0157] (e) an energy storage member unit comprising a plurality of energy storage members that have a length, which is a longest dimension of the energy storage member, and in use the length extends in a direction of the handle axis,

wherein, in use, the longitudinal axis intersects the energy storage member unit and the motor and fan assembly is located rearward of at least one of the energy storage members.

[0158] 19. The hand vacuum cleaner of clause 18 wherein, in use, a portion of the energy storage member unit is positioned in the main body.

[0159] 20. The hand vacuum cleaner of clause 18 wherein, in use, a portion of the energy storage member unit is positioned rearward of a pre-motor filter.

Clause Set 2

[0160] A hand vacuum cleaner comprising: [0161] (a) an airflow path from a dirty air inlet positioned at a front end of the hand vacuum cleaner to a clean air outlet positioned rearward of the dirty air inlet; [0162] (b) an air treatment assembly comprising an air treatment chamber, the air treatment chamber positioned in the airflow path, the air treatment chamber has an air treatment chamber air inlet, an air treatment chamber air outlet, a front end, a rear end and a centrally positioned longitudinal axis extending between the front and rear ends of the air treatment chamber; [0163] (c) a main body housing a motor and fan assembly that is positioned in the air flow path, the motor and fan assembly having a motor axis of rotation; [0164] (d) a handle extending away from the main body; and, [0165] (e) an energy storage member unit that is removably insertable into the handle.

[0166] 2. The hand vacuum cleaner of clause 1 wherein the handle has a first end provided on the main body, an opposed second end, a hand grip portion positioned between the first end and the opposed second end of the handle and the energy storage member unit is removably insertable through the second end of the handle.

[0167] 3. The hand vacuum cleaner of clause 1 wherein, when inserted, the energy storage member unit is recessed inwardly into the handle from the second end of the handle.

[0168] 4. The hand vacuum cleaner of clause 1 wherein the handle is a pistol grip handle.

[0169] 5. The hand vacuum cleaner of clause 1 wherein, in use, a portion of the energy storage member unit is positioned in the main body.

[0170] 6. The hand vacuum cleaner of clause 5 wherein, in use, the portion of the energy storage member unit that is positioned in the main body is located forward of the motor and fan assembly.

[0171] 7. The hand vacuum cleaner of clause 6 wherein, in use, the portion of the energy storage member unit that is positioned in the main body is located forward of a pre-motor filter.

[0172] 8. The hand vacuum cleaner of clause 6 wherein, in use, the portion of the energy storage member unit that is positioned in the main body is located rearward of a pre-motor filter.

[0173] 9. The hand vacuum cleaner of clause 6 wherein, in use, the portion of the energy storage member unit that is positioned in the main body is provided in the air flow path.

[0174] 10. The hand vacuum cleaner of clause 1 wherein the handle has a first end, an opposed second end, a hand grip portion and a handle axis that extends through the hand grip portion from the first end to the opposed second end, the energy storage member unit comprises a plurality of energy storage members that, in use, are axially arranged in a direction of the handle axis.

[0175] 11. The hand vacuum cleaner of clause 1 wherein the handle has a first end, an opposed second end, a hand grip portion and a handle axis that extends through the hand grip portion from the first end to the opposed second end, the energy storage member unit comprises a plurality of

energy storage members that have a length, which is a longest dimension of the energy storage member, and, in use, the length extends in a direction of the handle axis.

[0176] 12. The hand vacuum cleaner of clause 1 wherein the handle has a first end, an opposed second end, a hand grip portion and a handle axis that extends through the hand grip portion from the first end to the opposed second end, the energy storage member unit comprises a plurality of batteries that have a positive end and a negative end, a battery axis extends between the positive and negative ends and, in use, the battery axis extends in a common direction as the handle axis.

[0177] 13. The hand vacuum cleaner of clause 1 wherein the handle is removably mounted to a remainder of the hand vacuum cleaner.

[0178] 14. The hand vacuum cleaner of clause 13 wherein the handle is removably mounted to the main body.

[0179] 15. The hand vacuum cleaner of clause 13 wherein the handle is removable with the energy storage member unit.

[0180] 16. The hand vacuum cleaner of clause 1 wherein the handle comprises a stand for the hand vacuum cleaner.

[0181] 17. The hand vacuum cleaner of claim 1 wherein the second end of the handle comprises a stand for the hand vacuum cleaner.

[0182] 18. The hand vacuum cleaner of clause 1 wherein the second end of the handle is a stand for the hand vacuum cleaner.

[0183] 19. The hand vacuum cleaner of clause 1 wherein the hand vacuum cleaner has a first rearwardly extending side on which the handle is provided and an opposed second rearwardly extending side.

[0184] 20. The hand vacuum cleaner of clause 1 wherein, in use, the energy storage member unit extends from the first rearwardly extending side to the opposed second rearwardly extending side.

Claims

1. A hand vacuum cleaner comprising: (a) an airflow path from a dirty air inlet positioned at a front end of the hand vacuum cleaner to a clean air outlet positioned rearward of the dirty air inlet; (b) an air treatment assembly comprising an air treatment chamber, the air treatment chamber positioned in the airflow path, the air treatment chamber has an air treatment chamber air inlet, an air treatment chamber air outlet, a front end, a rear end and a centrally positioned longitudinal axis extending between the front and rear ends of the air treatment chamber; (c) a main body housing a motor and fan assembly that is positioned in the air flow path, the motor and fan assembly having a motor axis of rotation; (d) a handle extending away from the main body; and, (e) a unitary energy storage housing, wherein a first portion of the energy storage housing is provided in the handle and a second portion of the energy storage housing is provided in the main body, wherein, in use, the second portion of the energy storage housing, which is provided in the main body, is positioned forward of the motor and fan assembly.

2. The hand vacuum cleaner of claim 1 wherein the handle has a first end, an opposed second end, a hand grip portion and a handle axis that extends through the hand grip portion from the first end to the opposed second end and, in use, the first portion of the energy storage housing is provided in the hand grip portion.

3. The hand vacuum cleaner of claim 2 wherein the handle is a pistol grip handle.

4. The hand vacuum cleaner of claim 1 wherein the handle has a first end, an opposed second end, a hand grip portion and a handle axis that extends through the hand grip portion from the first end to the opposed second end, the energy storage housing comprises a plurality of energy storage members that, in use, are axially arranged in a direction of the handle axis.

5. The hand vacuum cleaner of claim 1 wherein the handle has a first end, an opposed second end, a hand grip portion and a handle axis that extends through the hand grip portion from the first end

to the opposed second end, the energy storage housing comprises a plurality of energy storage members that have a length, which is a longest dimension of the energy storage member, and, in use, the length extends in a direction of the handle axis.

6. The hand vacuum cleaner of claim 1 wherein the handle has a first end, an opposed second end, a hand grip portion and a handle axis that extends through the hand grip portion from the first end to the opposed second end, the energy storage housing comprises a plurality of batteries that have a positive end and a negative end, a battery axis extends between the positive and negative ends and, in use, the battery axis extends in a common direction as the handle axis.

7. The hand vacuum cleaner of claim 1 wherein, in use, the energy storage housing is positioned in the air flow path.

8. (canceled)

9. The hand vacuum cleaner of claim 1 wherein the main body further comprises a pre-motor filter and, in use, the second portion of the energy storage housing is positioned forward of the pre-motor filter.

10. The hand vacuum cleaner of claim 1 wherein the main body further comprises a pre-motor filter and, in use, the second portion of the energy storage housing is positioned rearward of the pre-motor filter.

11. The hand vacuum cleaner of claim 1 wherein the energy storage housing is removable from the hand vacuum cleaner.

12. The hand vacuum cleaner of claim 1 wherein the energy storage housing is removably insertable into the handle.

13. The hand vacuum cleaner of claim 1 wherein the handle has a first end provided on the main body, an opposed second end and a hand grip portion positioned between the first end and the opposed second end of the handle, and the energy storage housing is insertable into the second end of the handle.

14. The hand vacuum cleaner of claim 1 wherein, in use, the motor axis of rotation intersects the second portion of the energy storage housing.

15. A hand vacuum cleaner comprising: (a) an airflow path from a dirty air inlet positioned at a front end of the hand vacuum cleaner to a clean air outlet positioned rearward of the dirty air inlet; (b) an air treatment assembly comprising an air treatment chamber, the air treatment chamber positioned in the airflow path, the air treatment chamber has an air treatment chamber air inlet, an air treatment chamber air outlet, a front end, a rear end and a centrally positioned longitudinal axis extending between the front and rear ends of the air treatment chamber; (c) a main body housing a motor and fan assembly that is positioned in the air flow path, the motor and fan assembly having a motor axis of rotation; (d) a handle having a first end, an opposed second end, a hand grip portion positioned between the first end and the opposed second end of the handle and a handle axis that extends through the hand grip portion from the first end to the opposed second end; and, (e) an energy storage housing comprising a plurality of energy storage members that have a length, which is a longest dimension of the energy storage member, and in use the length extends in a direction of the handle axis, wherein, in use, the motor axis of rotation intersects the energy storage housing and the motor and fan assembly is located rearward of at least one of the energy storage members, and wherein, in use, the centrally positioned longitudinal axis intersects the energy storage housing.

16. The hand vacuum cleaner of claim 15 wherein, in use, a portion of the energy storage housing is positioned in the main body.

17. The hand vacuum cleaner of claim 15 wherein, in use, a portion of the energy storage housing is positioned rearward of a pre-motor filter.

18. A hand vacuum cleaner comprising: (a) an airflow path from a dirty air inlet positioned at a front end of the hand vacuum cleaner to a clean air outlet positioned rearward of the dirty air inlet; (b) an air treatment assembly comprising an air treatment chamber, the air treatment chamber positioned in the airflow path, the air treatment chamber has an air treatment chamber air inlet, an

air treatment chamber air outlet, a front end, a rear end and a longitudinal axis that extends between the front and rear ends of the air treatment chamber; (c) a main body housing a motor and fan assembly that is positioned in the air flow path, the motor and fan assembly having a motor axis of rotation; (d) a handle having a first end, an opposed second end, a hand grip portion positioned between the first end and the opposed second end of the handle and a handle axis that extends through the hand grip portion from the first end to the opposed second end; and, (e) an energy storage housing comprising a plurality of energy storage members that have a length, which is a longest dimension of the energy storage member, and in use the length extends in a direction of the handle axis, wherein, in use, the longitudinal axis intersects the energy storage housing and the motor and fan assembly is located rearward of at least one of the energy storage members.

19. The hand vacuum cleaner of claim 18 wherein, in use, a portion of the energy storage housing is positioned in the main body.

20. The hand vacuum cleaner of claim 18 wherein, in use, a portion of the energy storage housing is positioned rearward of a pre-motor filter.

21. The hand vacuum cleaner of claim 1 wherein a rearward projection of the rear end of the air treatment chamber extends, in a direction parallel to the centrally positioned longitudinal axis, through the energy storage housing prior to extending through the motor and fan assembly
